CORE MATH SYMBOLS
Summation ∑: Adds values in a sequence
Product ∏: Multiplies values in a sequence
Norm
LOGIC OPERATORS
And
Or
Not
Implies
If and Only If
PROBABILITY NOTATION
Probability of Event A
Joint Probability of A and B
Conditional Probability
Proportional To
Axiomatic Properties

Joint Distribution Notation
Marginalization
Normalization Constant
Distributions for Machine Learning
Entropy
Bayes Theorem
VECTOR & MATRIX OPERATIONS
Dot Product
Matrix Multiplication
Gradient Updates in Neural Network
SET NOTATION
Intersection
Union
Subset
Size of Set

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Bayesian Network Structure & D-Separation	**HMM & Particle Filtering**	**HMM Filtering**	**Naïve Bayes Classifier Network Structure**	**Linear Perceptron & Logistic Regression Decision Boundaries**
Gradient Descent	**Feed-Forward Neural Network**	**Confusion Matrix**	**Backpropagation Flow**	**Decision Boundary Comparison**

Bayesian Network Approximate Inference

Steps:

1. **Draw the Network:**
   - Verify the Bayesian Network is a Directed Acyclic Graph (DAG).
   - Check for no loops and label nodes with Conditional Probability Tables (CPTs).
2. **Topological Order:**
   - Arrange variables so parents precede children (use topological sort).
   - Example: Add nodes with no parents, remove them, and repeat.
3. **Prior Sampling:**
   - Formula: Sample each variable X using $P(X|\text{Parents}(X))$.
   - Steps: Follow topological order, sampling nodes based on their CPTs.
4. **Rejection Sampling:**
   - Formula: $P(Q|E) = \frac{\text{Count}(Q,E)}{\text{Count}(E)}$.
   - Steps: Generate samples in topological order, discard samples inconsistent with evidence E, and normalize retained samples.
5. **Gibbs Sampling:**
   - Formula: $P(X_i|\text{Markov Blanket}) \propto P(X_i|\text{Parents}) \prod_{\text{Children}} P(\text{Child}|X_i, \text{Child's Parents})$.
   - Steps: Initialize hidden variables randomly, fix evidence variables, resample each hidden variable using its Markov Blanket, and repeat until convergence.

Neural Networks

Steps:

1. **Forward Propagation:**
   - Formulas:
     - Pre-activation: $z^{[l]} = W^{[l]}a^{[l-1]} + b^{[l]}$.
     - Activation: $a^{[l]} = g(z^{[l]})$.
2. **Backward Propagation:**
   - Formulas:
     - Output Layer: $\delta^{[L]} = \nabla_a L \cdot g'(z^{[L]})$.
     - Hidden Layers: $\delta^{[l]} = W^{[l+1]^T}\delta^{[l+1]} \cdot g'(z^{[l]})$.
3. **Parameter Updates:**
   - Formulas:
     - $W^{[l]} \leftarrow W^{[l]} - \eta\nabla W^{[l]}$.
     - $b^{[l]} \leftarrow b^{[l]} - \eta\nabla b^{[l]}$.

Logistic Regression

Steps:

1. **Forward Pass:**
   - Formulas:
     - $z = w \cdot x + b$.
     - $\sigma(z) = \frac{1}{1+e^{-z}}$.
2. **Cross-Entropy Loss:**
   - Formula: $L = -\sum y\log(\hat{y}) + (1-y)\log(1-\hat{y})$.
3. **Gradient Descent:**
   - Formulas:
     - $\nabla w = (\hat{y} - y)x$, $w \leftarrow w - \eta\nabla w$.
     - $b \leftarrow b - \eta(\hat{y} - y)$.

Linear Perceptron

Steps:

1. **Initialize Parameters:** Set weights $w = 0$ and bias $b = 0$.
2. **Activation Function:**
   - Formula: $a = w \cdot x + b$.
3. **Prediction Rule:**
   - Formula: $\hat{y} = \text{sign}(a)$.
4. **Weight Update Rule:**
   - Formula: $w \leftarrow w + yx$, $b \leftarrow b + y$.
   - Steps: Update weights and bias if $\hat{y} \neq y$.

Naïve Bayes Classifier

Steps:

1. **Training Phase:**
   - Formulas:
     - Class Priors: $P(y) = \frac{\text{Count}(y)}{N}$.
     - Feature Likelihoods: $P(x_i|y) = \frac{\text{Count}(x_i,y)}{\text{Count}(y)}$.
     - Smoothed Likelihoods: $P(x_i|y) = \frac{\text{Count}(x_i,y)+\alpha}{\text{Count}(y)+\alpha|X|}$.
2. **Prediction Phase:**
   - Formula: $P(y|x_1, x_2, \ldots, x_n) \propto P(y)\prod P(x_i|y)$.
   - Steps: Compute $\log P(y) + \sum \log P(x_i|y)$ for each class, and select the class with the highest score.

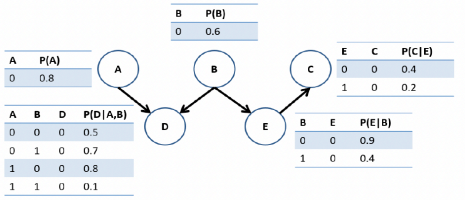
Hidden Markov Models

Steps:

1. **Define Components:**
   - Hidden States (X_t), Transition Probabilities ($P(X_t|X_{t-1})$), and Emission Probabilities ($P(E_t|X_t)$).
2. **Forward Algorithm:**
   - Formulas:
     - Initialization: $\alpha_1(x) = P(x)P(e_1|x)$.
     - Recursion: $\alpha_t(x) = P(e_t|x)\sum_{x'} P(x|x')\alpha_{t-1}(x')$.
   - Steps: Compute $\alpha_t(x)$ recursively for each time step, normalizing at every step.
3. **Particle Filtering:**
   - Steps:
     1. Initialize N particles distributed across states.
     2. Transition particles using $P(X_t|X_{t-1})$.
     3. Weight particles by $P(E_t|X_t)$.
     4. Resample particles based on weights for the next step.

Problem 1 – Bayesian Network Approximate Inference

Consider the following Bayesian Network, and use it to answer the questions that follow:



B	P(B)
0	0.6

A	P(A)
0	0.8

E	C	P(C|E)
0	0	0.4
1	0	0.2

A	B	D	P(D|A,B)
0	0	0	0.5
0	1	0	0.7
1	0	0	0.8
1	1	0	0.1

B	E	P(E|B)
0	0	0.9
1	0	0.4

1.1. Provide a *possible* order in which variables would be sampled by *Prior Sampling*:

[A, B, D, E, C]
(among other possibilities, as long as parents are always sampled before their children)

Suppose now that Prior Sampling is used to generate the following set of 10 samples.

Sample	A	B	C	D	E
0	0	1	1	0	0
1	0	0	1	1	0
2	1	0	0	1	1
3	0	1	1	1	1
4	1	1	1	0	0
5	0	0	0	1	1
6	0	1	1	1	0
7	1	0	0	1	0
8	0	0	1	1	0
9	0	0	1	1	0

1.2. Using these samples, find $\hat{P}(C = 1|D = 1)$

Find the proportion of C = 1 samples among the total of those for which D = 1:

$$\hat{P}(C = 1|D = 1) = \frac{count(C = 1, D = 1)}{count(D = 1)} = \frac{5}{8}$$

1.3. Using the same model, let's trace the generation of Gibbs Sampling in order to estimate $\hat{P}(B|C = 1, D = 1, E = 0)$. Suppose for the current sample we are sampling variables in the order of: [A, B, ...] and begin with the following random state. For each variable sampled thereafter, assume (for this exercise) that the most likely value of the distribution sampled-from is the one actually picked.

A	B	C	D	E
0	1	1	1	0

Sampling A (Provided as an Example)

1.3.1. From what distribution will A be sampled? Show the likelihoods of sampling each value.

Distribution Sampled: $P(A|B = 1, C = 1, D = 1, E = 0)$

$$P(A = 0|B = 1, C = 1, D = 1, E = 0) = \frac{P(A = 0)P(D = 1|A = 0, B = 1)}{\sum_a P(A = a)P(D = 1|A = a, B = 1)} = \frac{0.8 * 0.3}{0.8 * 0.3 + 0.2 * 0.9} \approx 0.57$$

1.3.2. Assuming our random number generator selects the value for A that is most likely, what will be the result of the above sample, and thus, the next state? (Fill in the value for A below):

A	B	C	D	E
0	1	1	1	0

Sampling B (Your Turn)

1.3.3. From what distribution will B be sampled? Show the likelihoods of sampling each value.

Distribution Sampled: $P(B|A = 0, C = 1, D = 1, E = 0)$

$$P(B = 0|A = 0, C = 1, D = 1, E = 0) = \frac{P(B = 0)P(D = 1|A = 0, B = 0)P(E = 0|B = 0)}{\sum_b P(B = b)P(D = 1|A = 0, B = b)P(E = 0|B = b)} = \frac{0.6 * 0.5 * 0.9}{0.6 * 0.5 * 0.9 + 0.4 * 0.3 * 0.4} \approx 0.85$$

1.3.4. Assuming our random number generator selects the value for B that is most likely, what will be the result of the above sample, and thus, the next state? (Fill in the value for B below):

A	B	C	D	E
0	0	1	1	0

Problem 2 – Hidden Markov Models

In bioinformatics, analyses of nucleotide sequences in DNA processing is a central, but noisy, process; determining the identity of each nucleotide (from the 4 possible: ACGT) is not only imperfect, but many nucleotides in the sequence are "non-coding," meaning they do not encode any protein sequences (and are, for many applications, ignorable). It turns out the likelihood of producing certain nucleotides (viz., GC) increases during "coding" ($X = 1$) sections, while the likelihoods of these two to express *decrease* during non-coding ($X = 0$). From lab data, we have parameterized the following HMM describing this process:

Initial

X	$P(X_0)$
0	0.50

Transitions

X_t	X_{t-1}	$P(X_{t+1}\|X_t)$
0	0	0.60
1	0	0.50

Emissions

E_t	$P(E_t\|X_t = 0)$	$P(E_t\|X_t = 1)$
A	0.30	0.20
C	0.20	0.30
G	0.20	0.30
T	0.30	0.20

2.1. Suppose we observe a (small) sequence of nucleotides: GGC. Treating our time step t as position within the sequence, use filtering to compute $B(X_3)$, our belief of the current state after observing the entire sequence, starting with the following as given:

	$t = 0$ *Init*	$t = 1$ $E_1 = G$	$t = 2$ $E_2 = G$	$t = 3$ $E_3 = C$
$B(X_t = 0)$	0.5	0.4	0.44	0.443
$B(X_t = 1)$	0.5	0.6	0.56	0.557

Forward Algorithm (Combining both Updates for Time and Observation):

$$B(X_3 = 0) = \alpha\, P(E_3 = C|X_3 = 0)\sum_{x_2} P(X_3 = 0|X_2 = x_2)B(x_2)$$
$$= \alpha\, P(E_3 = C|X_3 = 0)[P(X_3 = 0|X_2 = 0)B(X_2 = 0) + P(X_3 = 0|X_2 = 1)B(X_2 = 1)]$$
$$= \alpha\, 0.2 * [0.6 * 0.44 + 0.5 * 0.56] = \alpha\, 0.1088$$

$$B(X_3 = 1) = \alpha\, P(E_3 = C|X_3 = 1)[P(X_3 = 1|X_2 = 0)B(X_3 = 0) + P(X_3 = 1|X_2 = 1)B(X_2 = 1)]$$
$$= \alpha\, 0.3 * [0.4 * 0.44 + 0.5 * 0.56] = \alpha\, 0.1368$$

$$\alpha = \frac{1}{0.1088 + 0.1368} = \frac{1}{0.2456}$$

$$B(X_3 = 0) = \frac{0.1088}{0.2456} \approx 0.443; \therefore B(X_3 = 1) = 1 - B(X_3 = 0) \approx 0.557$$

In this setting, since we really care about the *most likely state* at any given part of the sequence, it's often useful to seek an **explanation** of what led us to witness the evidence at any given time. However, to compute this *most likely explanation*, we need to amend our Forward Algorithm slightly; instead of summing over all *possible* previous states in the transitions, we focus on only the most likely, labeled $M(X_t)$, in what is known as the **Viterbi Algorithm**.

Forward Algorithm	*Viterbi Algorithm*
$B(X_{t+1}) = \alpha\, P(E_{t+1}\|X_{t+1})\sum_{x_t} P(X_{t+1}\|x_t)B(x_t)$	$M(X_{t+1}) = \alpha\, P(E_{t+1}\|X_{t+1})\max_{x_t} P(X_{t+1}\|x_t)M(x_t)$

2.2. Suppose we observe the sequence $GGCA$ and are using the Viterbi Algorithm to find the most likely explanation of the state at each point in its generation. Using the following partially-completed computations, find $M(X_4)$: (hint: it *is* as simple as it seems, that's why it's nice).

	$t = 0$ *Init*	$t = 1$ $E_1 = G$	$t = 2$ $E_2 = G$	$t = 3$ $E_3 = C$	$t = 4$ $E_4 = A$
$M(X_t = 0)$	0.5	0.4	0.4	0.4	0.6
$M(X_t = 1)$	0.5	0.6	0.6	0.6	0.4

$$M(X_4 = 0) = \alpha\, P(E_4 = A|X_4 = 0)\max_{x_3} P(X_4 = 0|X_3 = x_3)\, M(X_3 = x_3)$$
$$= \alpha\, 0.3\max(0.6 * 0.4, 0.5 * 0.6)$$
$$= \alpha\, 0.09$$

$$M(X_4 = 1) = \alpha\, P(E_4 = A|X_4 = 1)\max_{x_3} P(X_4 = 1|X_3 = x_3)\, M(X_3 = x_3)$$
$$= \alpha\, 0.2\max(0.4 * 0.4, 0.5 * 0.6)$$
$$= \alpha\, 0.06$$

$$\therefore \alpha = \frac{1}{0.15}$$
$$\therefore M(X_4 = 0) = 0.6;\ M(X_4 = 1) = 0.4$$

2.3. Examining the completed table above, determine the *explanation* (i.e., the most likely value of the state) for the observed sequence in all steps $t = 1 \to 4$:

	$t = 1$ $E_1 = G$	$t = 2$ $E_2 = G$	$t = 3$ $E_3 = C$	$t = 4$ $E_3 = A$
X_t^*	$X = 1$	$X = 1$	$X = 1$	$X = 0$

1.2. Using the data from the previous section, suppose we instead *smooth* our estimates for the feature likelihoods using Laplacian Smoothing with $k = 10$. Compute $P_{LAP,10}(C = 0|S = 1)$

$$P_{LAP,10}(C = 0|S = 1) = \frac{c(C = 0, S = 1) + k}{c(S = 1) + k|X|} = \frac{40 + 10}{400 + 10 * 2} \approx 0.12$$

[|] Note: |X| = 2 here because the variable to which we are artificially adding samples is C, which is binary. Since we're adding $k = 10$ samples to each of $C = 0,1$, that's why we add 20 extra to the total in the denominator!

1.3: Using the NBC you learned in Part 1.1, determine (showing your work) the most likely class for each of the following data points.

1. $\{A = 0, B = 1, C = 0\}$

$$P(S = 0|A = 0, B = 1, C = 0) \propto P(S = 0)P(A = 0|S = 0)P(B = 1|S = 0)P(C = 0|S = 0)$$
$$= 0.6 * 0.4 * 0.2 * 0.25$$
$$= 0.012$$

$$P(S = 1|A = 0, B = 1, C = 0) \propto P(S = 1)P(A = 0|S = 1)P(B = 1|S = 1)P(C = 0|S = 1)$$
$$= 0.4 * 0.7 * 0.25 * 0.1$$
$$= 0.007 \text{ (in Die Another Bayes)}$$

$$P(S = 0|A = 0, B = 1, C = 0) > P(S = 1|A = 0, B = 1, C = 0)$$
$$\therefore \textit{Choose } S = 0$$

2. $\{A = 1, C = 1\}$ [Hint: Remember that NBCs are still Bayesian Networks, so B as a missing feature shouldn't be a problem – make sure your answer derives why]

$$P(S|A, C) \propto \sum_b P(S)P(A|S)P(b|S)P(C|S) = P(S)P(A|S)P(C|S)\sum_b P(b|S) = P(S)P(A|S)P(C|S)$$
$$P(S = 0|A = 1, C = 1) \propto P(S = 0)P(A = 1|S = 0)P(C = 1|S = 0)$$
$$= 0.6 * 0.6 * 0.75$$
$$= 0.270$$

$$P(S = 1|A = 1, C = 1) \propto P(S = 1)P(A = 1|S = 1)P(C = 1|S = 1)$$
$$= 0.4 * 0.3 * 0.9$$
$$= 0.108$$

$$P(S = 0|A = 1, C = 1) > P(S = 1|A = 1, C = 1)$$
$$\therefore \textit{Choose } S = 0$$

3.2. Suppose our sample $x^{(i)}$ was meant to have label $y^{(i)} = 0$. Determine the new weights that would be associated with each class in a single step of stochastic gradient descent with learning rate $\eta = 0.1$ and cross entropy loss. Show your work in the box below.

$$\frac{\delta L_{CE}}{\delta w_0} = -\left[1 - P\left(\widehat{y^{(i)}} = 0\middle|x^{(i)}; w\right)\right] * f(x)$$
$$= -[1 - 0.25] * \langle 1,1,1\rangle$$
$$= \langle -0.75, -0.75, -0.75\rangle$$

$$\frac{\delta L_{CE}}{\delta w_1} = -\left[0 - P\left(\widehat{y^{(i)}} = 0\middle|x^{(i)}; w\right)\right] * f(x)$$
$$= -[0 - 0.67] * \langle 1,1,1\rangle$$
$$= \langle 0.67, 0.67, 0.67\rangle$$

$$\frac{\delta L_{CE}}{\delta w_2} = -\left[0 - P\left(\widehat{y^{(i)}} = 2\middle|x^{(i)}; w\right)\right] * f(x)$$
$$= -[0 - 0.09] * \langle 1,1,1\rangle$$
$$= \langle 0.09, 0.09, 0.09\rangle$$

$$w_0 = w_0 - \eta\frac{\delta L_{CE}}{\delta w_0} = \langle 2,2,-3\rangle - 0.1\langle -0.75, -0.75, -0.75\rangle = \langle 2.075, 2.075, -2.925\rangle$$
$$w_1 = w_1 - \eta\frac{\delta L_{CE}}{\delta w_1} = \langle -1,1,2\rangle - 0.1\langle 0.67, 0.67, 0.67\rangle = \langle -1.067, 0.933, 1.933\rangle$$
$$w_2 = w_2 - \eta\frac{\delta L_{CE}}{\delta w_2} = \langle 2,-3,1\rangle - 0.1\langle 0.09, 0.09, 0.09\rangle = \langle 1.991, -3.009, 0.991\rangle$$

$w_0 =$	$w_1 =$	$w_2 =$
$\langle 2.075, 2.075, -2.925\rangle$	$\langle -1.067, 0.933, 1.933\rangle$	$\langle 1.991, -3.009, 0.991\rangle$